Sensors Lab Course

Gyroscope, accelerometer, pressure sensor, magnetometer, gas sensor

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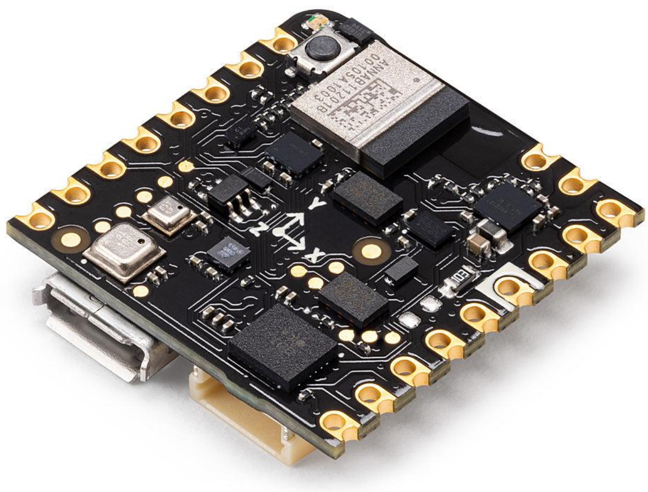
Winter term 2025/26



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Introduction

This lab course is a mandatory part of the lectures *Sensors* and *Sensorik* from Master of Science programs in *Microsystems Engineering*, *Mikrosystemtechnik*, and *Embedded Systems Engineering* at the Faculty of Engineering. Through the three modules of the lab course, we provide you with hands-on experience to deepen the knowledge from the lectures. We would like you to spark your interest in sensor applications playfully but simultaneously expect a university-level performance during the experimentation and while writing your reports.

All experiments use the Arduino Nicla Sense ME platform (Figure 1), which comprises four state-of-the-art integrated sensors from Bosch Sensortec, an Arm Cortex M4 processor, and Bluetooth connectivity. The different sensors are an inertial measurement unit (IMU) measuring acceleration and rotation (BHI260AP), a pressure sensor (BMP390), a magnetometer (BMM150), and a gas sensor (BME688) providing deduced parameters like equivalent CO_2 and volatile organic compounds (VOC) concentrations together with temperature and humidity.

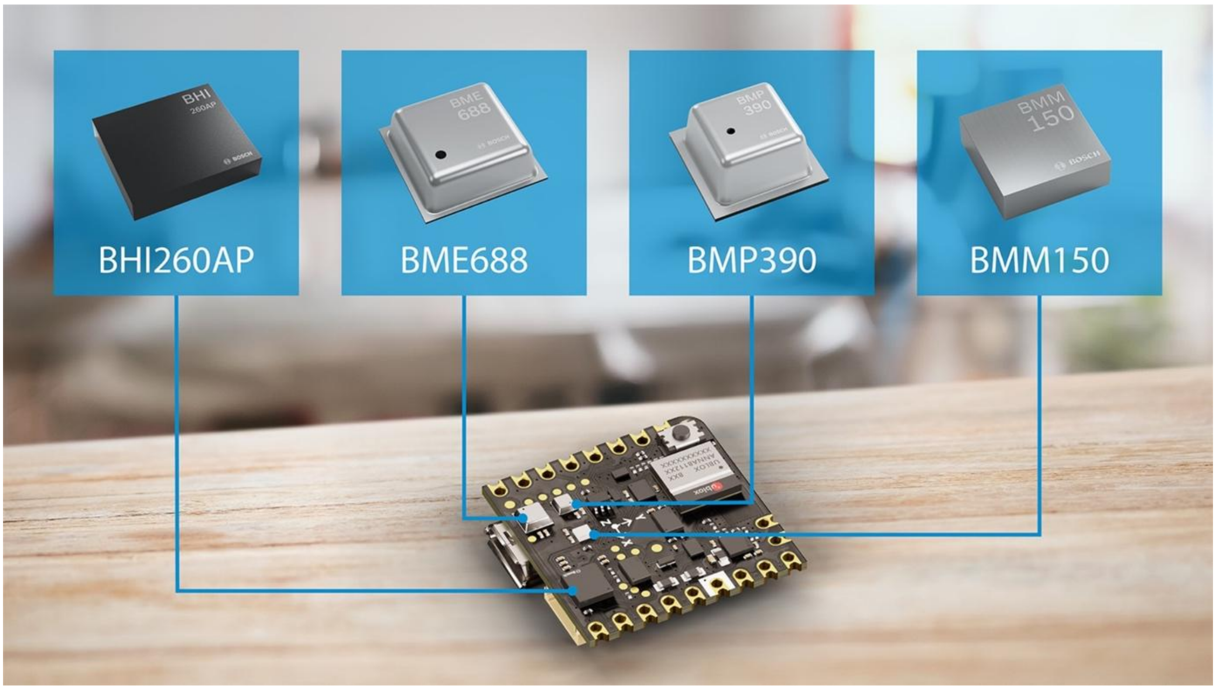


Figure 1: The Arduino Nicla Sense ME board comprises several sensors and connectivity modules. It can be programmed with the Arduino IDE using a USB connection.
Image: Bosch Sensortec.

The lab course consists of self-learning modules. Each student borrows one Nicla Sense ME board from the faculty library.[1]. Nothing else is needed besides a computer with a USB port, preferably a notebook. Within the given schedule (i.e., deadlines for report submission), you can work on the experiments at your own pace.

While there will be no physical meetings, please stay connected with your colleagues and the tutors through the University's online learning platform[2] (ILIAS). **You can ask questions at any time in the forum in the ILIAS group of the lab course.** We will help you there promptly,

[1] Address and opening hours: https://www.ub.uni-freiburg.de/?id=3281
[2] https://ilias.uni-freiburg.de

and other students can suggest solutions. Please support each other – in the lab course, you are not in competition with each other but should all explore the exciting world of sensors together. If you are new to the University, you can also use the forum to find mates to work together.

Learning Objectives

We want you to reach the learning objectives by the end of the lab course.

1. You have practical experience with different state-of-the-art sensors (accelerometer, gyroscope, pressure sensor, magnetometer, gas sensor, humidity sensor, temperature sensor) and an embedded sensor platform.
2. You can program an embedded system to interface with different sensors and provide the data to a connected computer.
3. You know how to perform sensor measurements according to scientific standards.
4. You can analyze sensor data (filtering, integration, differentiation).
5. You can document and appropriately discuss your measurements in a report.
6. You understand the working principles of the different sensors and relate your measurements to the limitations of the sensor principle.

Formalities

The lab course accounts for 2 ECTS, translating into a workload of 60 hours over the semester, on average, 4 hours per week during lecture time.

To pass the course, you need all three reports to be acknowledged. There is a deadline for each report. The reports must be submitted as a PDF file. The used microcontroller code needs to be submitted as a zip file. You will find an upload option for these files in the ILIAS course of this lab course.

You can rework a report and resubmit within one week if a report needs to be revised. In total, you have three options to rework. Please be aware: if you need all three rework options to pass the first, you must succeed with the other two reports in the first submission.

We encourage you to do the practical experiments with your colleagues (small groups of up to 4 students). **However, each student must write the report, make the plots, and discuss the results individually.** When you share code or perform experiments in a group resulting in the same data, you must indicate the code or data in your report and declare with whom you collaborated.

Before the three modules, we reserved additional time for an onboarding phase to familiarize you with the software and hardware platform. No report from your side is required here; instead, we provide a sample report reflecting this experiment and illustrating our expectations.

Disclosure

Bosch Sensortec kindly provided the Arduino Nicla Sense ME boards.

Schedule

We suggest the schedule below for working on the lab course. However, the lab course is self-paced; only the red-marked dates (submission deadlines for the report) are binding.

Date	**Task**
Week 1 (13.-17.10.25)	Getting started, collect the hardware from the faculty library
Week 2 (20.-25.10.25)	Module 0: Gyroscope and Acceleration Sensors (“onboarding”)
Week 3 (27.-30.10.25)	Module 0: Gyroscope and Acceleration Sensors (“onboarding”)
Week 4 (03.-07.11.25)	Module 1: Acceleration and Pressure Sensors
Week 5 (10.-14.11.25)	Module 1: Acceleration and Pressure Sensors
Week 6 (17.-21.11.25)	Module 1: Acceleration and Pressure Sensors
Week 7 (24.-28.11.25)	Module 1: Acceleration and Pressure Sensors
Friday, 28.11.25	Submission of the report to Module 1
Week 8 (01.-05.12.25)	Module 2: Magnetic Sensors
Week 9 (08.-12.12.25)	Module 2: Magnetic Sensors
Week 10 (15.-19.12.25)	Module 2: Magnetic Sensors
Week 11 (07.-09.01.26)	Module 2: Magnetic Sensors
Friday, 09.01.26	Submission of the report to Module 2
Week 12 (12.-16.01.26)	Module 3: Gas Sensors
Week 13 (19.-23.01.26)	Module 3: Gas Sensors
Week 14 (26.-30.01.26)	Module 3: Gas Sensors
Week 15 (02.-06.02.26)	Module 3: Gas Sensors
Friday, 06.02.26	Submission of the report to Module 3
Within 150 days	Return the hardware to the faculty library

Submission

The report must be submitted as a PDF file. The used microcontroller code needs to be submitted as a zip file. You will find an upload option for these files in the ILIAS course of the lab course. You must strictly comply with the following naming convention:

SensLab_M<*Module*>_<*Family name*>_<*Matriculation no*>[_rev<*revision*>].pdf or .zip

For example, Isaac Newton would have filed his report as SensLab_M1_Newton_3141592.pdf and the third revision as SensLab_M1_Newton_3141592_rev3.pdf (he would have struggled with today’s way of writing his laws …).

Arduino Nicla Sense ME platform

The *Arduino Nicla Sense ME* is a versatile development board with many sensors allowing simple programming using the Arduino Integrated Development Environment (IDE). At the same time, it is a platform enabling complex *artificial intelligence* (AI) applications with substantial computational demands. The platform allows for advanced *Internet of Things* (IoT) applications with data processing at the sensors (*edge computing*) before a transmission unit (e.g., an external WiFi module) transfers the processed information to the cloud. A built-in module features connectivity over Bluetooth Low Energy (BLE); a Universal Serial Bus (USB) allows cable-bound serial connection to a computer. The board design focuses on low power consumption with a supply from a battery but also allows powering from the USB.

Within the lab course, we focus on the board's standalone operation and programming with the Arduino IDE. We use the USB port for the power supply, programming, and data retrieval. You are encouraged to explore the other functionalities to deepen your skills beyond the mandatory tasks.

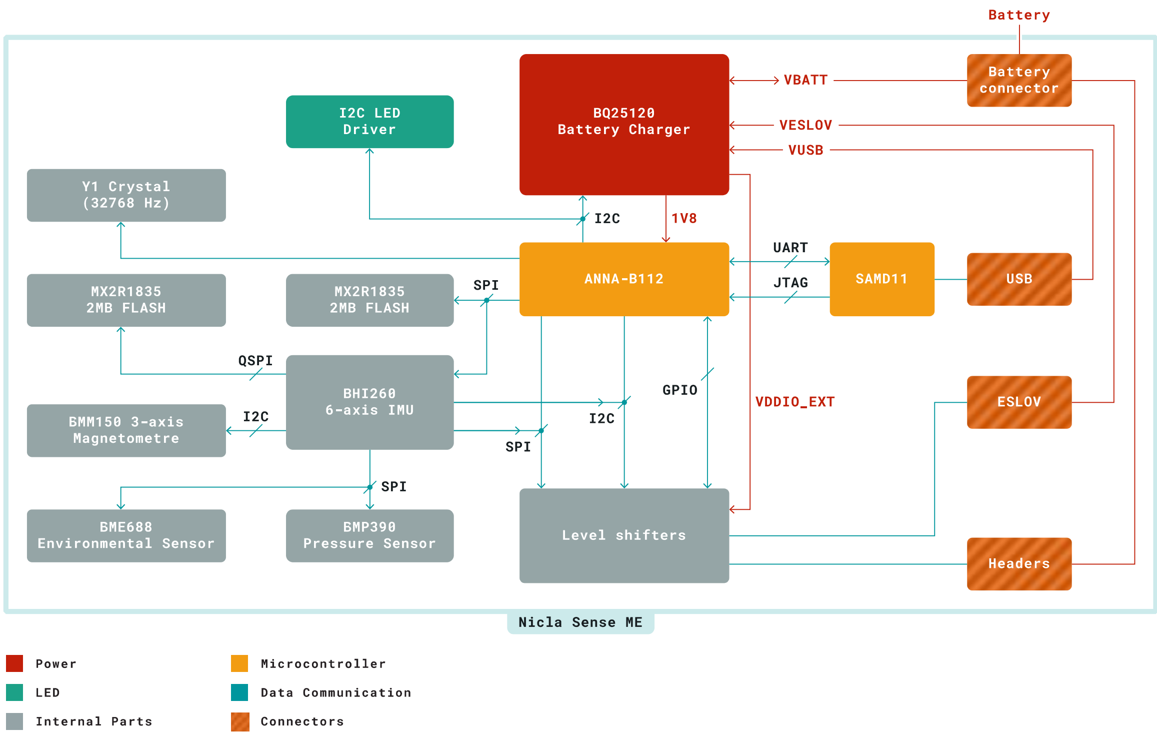


Figure 2: Nicla Sense ME block diagram [1]. The ANNA-B112 module hosts the main microcontroller. The magnetometer (BMM150), the gas sensor (BME688), and the pressure sensor (BMP390) connect to the inertial measurement unit (BHI260), which not only comprises the accelerometers and gyroscopes but also acts as a smart sensor hub with its processor enabling AI applications.

The ANNA-B112 Bluetooth module comprises the board's main microcontroller with an ARM Cortex M4 architecture and 512 kB flash memory for the program code. The block diagram (Figure 2) illustrates the other system components and reveals the sensor's connections. The different sensors connect to the BHI260 inertial measurement unit (IMU), which also contains a microcontroller acting as a sensor hub. The board's data sheet [1] provides a more detailed description of the system architecture and components.

We use the pre-installed firmware in the BHI260 for the lab course. The firmware provides interfaces called *virtual sensors*. A virtual sensor may represent the raw values of the sensor, calculate values based on calibration data, or consider other parameters (e.g., temperature) to provide compensated values. We focus on raw values in the lab course as much as possible, while professional applications prefer compensated data. Other virtual sensors provide fused data, e.g., for orientation or advanced features like gesture recognition. In the description of each module, we provide which virtual sensor you should use. Once more experienced, you can run an example program[3] that generates a list of all available virtual sensors.

Tools

A simple way is **to program the microcontroller** of the Nicla Sense ME with the *Arduino IDE*. An integrated development environment (IDE) combines an editor with a compiler and, if necessary, other tools. The Arduino IDE is available for Windows, Linux, and macOS and includes the GCC (GNU C compiler). Apart from the editor, the IDE contains some basic libraries and allows the installation of additional ones easily. Besides compiling, the IDE can transfer the program to the microcontroller's flash storage.

Furthermore, the IDE contains a console for the serial interface and a simple plot program for data received via the serial interface. The free Windows software *PuTTY*[4] or *HTerm*[5] are alternatives to the Arduino IDE’s serial console and facilitate saving the gathered data. If you are familiar with programming, use your preferred IDE, such as Microsoft Visual Studio Code, which provides Arduino support through extensions. We only support the Arduino IDE, PuTTY, and HTerm in the forum.

Module 0 guides you through the necessary online resources to get the Arduino IDE to run for programming the Nicla Sense ME.

[3] https://github.com/arduino/nicla-sense-me-fw/tree/main/Arduino_BHY2/examples/ShowSensorList, also available in the Arduino IDE under examples once the Arduino_BHY2 library is installed
[4] https://www.putty.org
[5] https://www.der-hammer.info/pages/terminal.html

Writing a Scientific Lab Report

This section introduces you to how to write a scientific lab report. The goal should be that another scientist with a similar background and the same equipment can reproduce your results. That means you must balance over-educating and skipping essential details, hiding how you did the experiments. Length is not a quality criterion. An excellent report is concise, which means as short as possible while mentioning everything necessary in the abovementioned sense. A good guideline is to imagine that you should be able to reproduce everything in three years without starting the investigations from scratch. We provide a sample report for Module 0 (the "onboarding") illustrating the most essential aspects.

Before Writing

Writing an excellent scientific lab report already starts before the experiment. Most importantly, carefully consider which parameters might influence your measurements and adjust accordingly. Try to figure out as much information as possible about your equipment, e.g., which principles the sensors rely on or which value range is to be expected. It is also helpful to look over the learning objectives of the student's lab to understand what we expect from you. Discussing with your colleagues helps you better understand the topic and shows you if you need more information in some parts.

While doing the experiments, you should document everything! Do not underestimate how fast one forgets the details. Don't count on being an exception to that rule. Reconstructing information later is a colossal waste of time. You may use your smartphone's camera to document the setup.

Content

While there are no internationally binding guidelines for writing a lab report, a de facto standard has evolved following the structure of research papers. The report should be structured as follows:

- The **title page** must indicate what kind of report it is, who wrote it (from which institution), and the date of the last update. Here, you need to state the lab course's name, the module, your name, your matriculation number, and the date. If you use data generated in a team, state your colleagues' names and matriculation numbers.
- The **Introduction** (*Einführung*) provides short background information and an experiment overview. A precise formulation of the experiments' goals helps the writer and the reader.
- You can keep the **Theory** (*Theorie*) section concise, assuming a generally educated engineer is the reader. Never recapitulate Newton's Laws in all detail, while more specific relationships like the Barometric Formula are worth mentioning. Use this section for the lab course to briefly explain the sensors' working principles theory.
- In the **Methods** (*Methoden*), you should describe the setup and procedures so that you can repeat the experiment in three years. Also, consider listing software versions if you

suspect they influence the measurement results. Any environmental conditions, such as temperature, exceptional weather conditions, etc., are essential.

- **Results and Discussion** (*Ergebnisse und Diskussion*) can be separated or combined into a single section, as it has become common in many scientific journals. Also, if combined, make the result (what you observed) and the discussion (your interpretation) clear while writing. The results (your observation) are always valid, while the discussion might get outdated when newer research results improve our understanding. That is why researchers in fundamental sciences prefer separate sections; a combination is the better choice for a lab report.
- The **Summary** (*Zusammenfassung*) wraps up the results and major findings. Keep it short and easy to read. Three to five sentences are an ideal guideline for the modules in this lab course.
- The **References** (*Quellenangaben*) contain all your sources (see below for details).

Language

You can choose between German or English (British or American spelling). Be consistent in your choice. That includes using the correct decimal separator (“.” vs. “,”) and using the same language for labeling your plots. Scientific language is simple. Focus on short and precise sentences. Avoid bloomy formulation or literary style that excludes non-native speakers. Please remember you are at a university, so write complete sentences and use a spell checker!

Quantities, Numbers, and Units

You must use SI units, i.e., no imperial units. SI stands for the *Système international d'unités*, the International System of Units:

- There are seven *base units*: m, kg, s, A, K, mol, and cd
- For convenience, many *derived units* are in use: rad, sr, Hz, N, Pa, J, W, C, V, F, Ohm (Ω), S, Wb, T, H, °C, lm, lx, Bq, Gy, Sv, and kat
- Expression of small or large quantities benefit from *prefixes*: Y (10^{24}), Z (10^{21}), E (10^{18}), P (10^{15}), T (10^{12}), G (10^{9}), M (10^{6}), k (10^{3}), h (10^{2}), da (10^{1}), d (10^{-1}), c (10^{-2}), m (10^{-3}), μ (10^{-6}), n (10^{-9}), p (10^{-12}), f (10^{-15}), a (10^{-18}), z (10^{-21}), and y (10^{-24})
- Possible *exceptions*, which are also accepted in this lab course, account for traditions in the different subjects: min, h, d, y, l (liter), g (gram), t (ton), dB, eV, Da, Å (10^{-10}), M (molar), ppm, and ppb

Take care to report only **significant digits**. You should calculate with full precision but finally round the results to meaningful digits. Never pretend to have higher precision in your results than your measurement system has!

Example: When the 13th person entered the elevator, there was an alert, and the display said, “Overweight: 973 kg”. What was the average mass m of the people in the elevator?

$m = \frac{973\text{ kg}}{13} = 74.8\cancel{4615}\text{ kg}$ … round to the significant digits (three in this case)

You can use the standard deviation for a measurement series with normally distributed data to judge the significant digits. If you know about the sensor's accuracy, consider this information when reporting results.

Example: If the above-mentioned elevator's balance has an accuracy of ±1 kg the average weight is $m = 75$ kg.

Typesetting of Equations and Units

Consistent typesetting of equations and units helps the reader oversee your report and shows that your work is professional. Use the italic font for your variables and the upright font for labels and units. Do not forget the blank between numbers and units (preferably a "protected space" in case your text processor supports it to avoid line breaks in between). Use the correct decimal separator ("." vs. ",") depending on your language.

Example: The length l of a marathon (indicated by the label "m") is $l_{\mathrm{m}} = 42.195$ km. Refer to different lengths by l_i $(i = 1, 2, 3)$.

Introduce each variable at its first occurrence. Sometimes, a list with all variables, especially in an interdisciplinary context, helps the reader.

Never use old-fashioned pseudo-code to typeset equations in a report.

Example: Use $P = I^2 R$ instead of $P = I\hat{}2 \ * R$.

Citations and References

Statements in the text, values, data, and images from another source require appropriate citation. You can cite another work within the text or in the caption of a figure by introducing a numeric or alphanumeric reference.

Examples:

- Citing a book is more manageable than traveling through the galaxy [1].
- Most microsystems engineering students have read Richard Feynman's famous lecture. The transcript can also be found in a journal publication [2].
- If you cite a web page like our faculty's [3], be sure to mention the date of last access.

In a section at the end of your report, you provide the source of your references. Footnotes are also sometimes used for the sources but are less common in the engineering disciplines.

Examples:

[1] D. Adams, The hitchhiker's guide to the galaxy. New York, NY, USA: Del Rey/Ballantine, 2005.

[2] R. Feynman, "There's plenty of room at the bottom", Journal of Microelectromechanical Systems, vol. 1, no. 1, pp. 60-66, 1992.

[3] “The Faculty of Engineering – Solutions for the future — Faculty of Engineering”, Tf.uni-freiburg.de, 2022. [Online]. Available: https://www.tf.uni-freiburg.de/en. [Accessed: 01-Aug-2022].

There are many citation styles; the above examples used the *IEEE style*. You can use any style typical in the engineering disciplines but ensure consistency. Citing whole sentences or paragraphs, common in the arts and humanities, is unusual in the engineering disciplines or natural sciences and should be avoided.

Make sure to distinguish the scientific correct referencing from copyright aspects. The latter is not an issue in an unpublished lab report, but copyright-free material also requires proper referencing. If you publish your lab report, ensure that it does not violate any copyright.

Plots

Scientific plotting is essential to report data accurately and display measurement results in a report. Students often need to pay more attention to correct and meaningful plots. Sometimes, even the axes have no labels, or units are missing. Please pay utmost attention to the graphical representation of your data while writing your report.

Figure 3 shows an example plot for two-dimensional data. Some rules and hints help you produce correct and meaningful diagrams:

- Each axis needs a label comprising the unit, and the scaling gets clear, which means you need at least two ticks to relate the data points to their numerical correspondence.
- Avoid the unit in rectangular brackets (“Length [m]”). In some engineering disciplines, the rectangular brackets are also used as an operator or, in analytical chemistry, represent the concentration.
- If a quantity is dimensionless, you can use “1” instead of the unit, or if the unit is device-specific, indicating an arbitrary unit by “a. u.” is possible.
- When important, scale to include the zero in your diagram but avoid the data sticking only to a small fraction of the diagram's area. Indicate the usage of nonlinear scaling (e.g., logarithmic scaling).
- If only a few discrete points are available, avoid connected lines. If there are many data points, you can use lines instead.
- Use legends to clarify your plots and labels (letters, arrows) to highlight specific sections you want to refer to in your discussion.
- Take care of color-blind people. A rough approach could be to check if the diagram would also work as a greyscale image. You may use different markers or line styles.

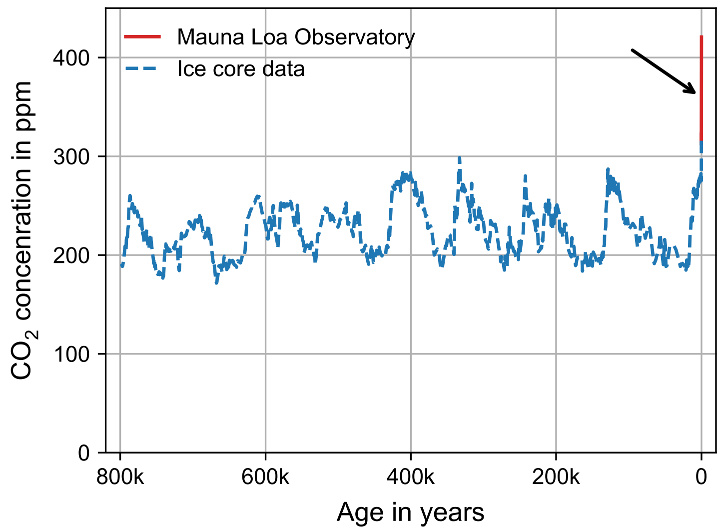


Figure 3: CO_2 concentration in the atmosphere: the Keeling curve (marked with the arrow), named after Charles David Keeling from the Scripps Institution of Oceanography, has been measured since 1958 at the Mauna Loa Observatory [2]. The curve is much more alarming in a larger context combined with paleoclimatic ice core data [3].

Plagiarism and Cheating

Plagiarism and cheating are the most critical frauds in science. Worst, reporting fake data hinders scientific progress, harms people, and affects science's reputation. That is why **we seriously pursue plagiarism and cheating in the lab course, i.e., you fail.** If you do that repeatedly, the University may exclude you from further studies. You all know it, but once again, using someone else's texts, images, code, data, solutions, etc., without giving the source is plagiarism. Sources include books, lab course manuals or lecture notes, webpages, and your colleagues. While we encourage group work for the experiments, ensure that you write your report on your own. You can use data measured as a group, but you must evaluate and plot it yourself. Never generate data you didn't measure; never use calculations or tools to eliminate unwanted data points without mentioning them. **If you get caught, you won't have saved any time!**

Tools

The essential tool for report writing is the **text processor**. Mainly, two possibilities exist: *Microsoft Word* (or its free pendants in *Open Office/LibreOffice*) and *LaTeX*. Word had a bad reputation, to some extent originating in the poor typesetting of equations in older versions and the possibility of writing and formatting texts in an unstructured way. If you strictly use style sheets (called "Styles" or "Formatvorlagen") for the formatting and resist all Word Art-like features, scientific writing is possible. The equation editor improved substantially in Office 2016, allowing some LaTeX code and related shortcuts to speed up writing equations. Students can access Microsoft Office 365 Pro Plus[6] for a small annual price. Microsoft Word runs on Windows and macOS, and the free pendants are on Linux as well. The alternative to Word is

[6] https://bildung365.de/

LaTeX, which features high-quality typesetting and facilitates the structuring of your document without the possibility of getting concerned about the final layout while writing. *Overleaf*[7] is a convenient online LaTeX environment that allows report writing in the cloud. The free plan is well-suited for a lab report. Generating PDF is easily possible in Word (save the document as PDF) or LaTeX (use pdflatex) without additional software. A warning, which is especially helpful for some LaTeX users: **Don't forget the content over the form!**

For **references** in a lab report, you could use manual formatting or free (advertisement-financed) web services, e.g., *Cite This For Me*[8]. You may also use a professional reference manager, for larger documents such as a thesis highly recommended. There are several options, e.g., *Mendeley*[9] which offers a free version, or *Citavi,* for which a campus license exists[10].

Often, **data processing** and **plotting** go hand in hand. *MATLAB* by MathWorks is a programming language and numeric computing environment well suited for signal processing, including plotting functionality. MATLAB is available for Windows, Linux, and macOS; there is a state license[11] that provides students with MATLAB for free. An alternative would be *Origin* by OriginLab, a Windows software dedicated to scientific plotting and data processing. Students can access OriginPro Campus license[12] for free. If you prefer plotting without writing code, Origin is a good choice. In contrast, Microsoft Excel is not well suited for scientific plotting (also, you might, with a lot of effort, extract similar-looking results). If you are into *Python* programming, libraries like *pandas* for data handling and *matplotlib* for plotting are also good choices to obtain publication-grade scientific plots efficiently. However, learning Python just for lab courses is overkill. In the forum, we provide support for MATLAB only.

Drawing software is less relevant than the previously described tools but still a necessity. Often, a block diagram or a sketch explains relationships much better than lengthy text. While professional tools like *Adobe Illustrator* are out of reach for students' budgets, free tools like *draw.io* are a good compromise between capability and learning effort for a lab report. The latter also provides various shapes for flow diagrams and elements for electrical circuits. Draw.io is available as a desktop application[13] or cloud service[14] to draw directly in a browser.

[7] https://www.overleaf.com
[8] https://www.citethisforme.com/
[9] https://www.mendeley.com/
[10] https://www.rz.uni-freiburg.de/en/services/procurement/software/citavi-en
[11] https://www.rz.uni-freiburg.de/en/services/procurement/software/matlab-license
[12] https://www.rz.uni-freiburg.de/en/services/procurement/software/info-originpro
[13] https://github.com/jgraph/drawio-desktop/releases
[14] https://app.diagrams.net

Module 0: Gyroscope and Acceleration Sensors (“onboarding”)

Do you remember riding a merry-go-round as a child? The action got bigger the further you leaned out, and the faster you spun. In this module, we will quantify our childhood remembering by measuring the rotation speed and the acceleration at different distances from the center of rotation. Within this module, you install the software to program the Nicla Sense ME on your computer and make your first steps in programming the device. You can follow the tasks as we did on a merry-go-round or simply turn around yourself with the sensor in your hand, comparing bent and extended arms.

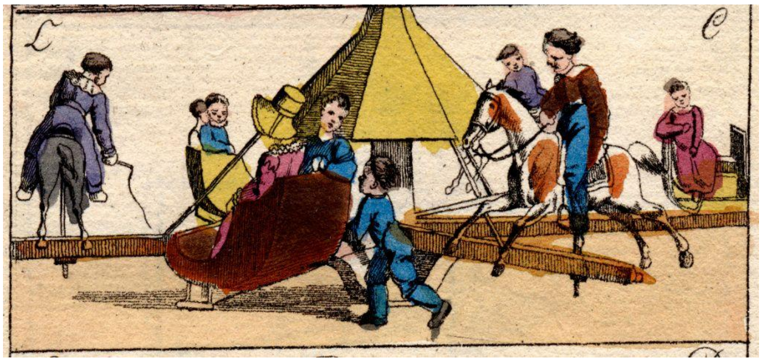


Figure 4: Merry-go-round, illustration dating back nearly two centuries. [4]

The primary intention of this module is to get you started with the sensor platform. We provide a sample report documenting our merry-go-round ride to illustrate our expectations for the other modules. Please note that the tasks are shorter than in modules 1 to 3 – so is the sample report.

Inertial Forces

Forces that appear when the frame of reference is non-inertial are *inertial forces*. Because of their dependency on the frame of reference, the term “fictitious force” is common, although the forces may have very real consequences (think of your ride on a merry-go-round).

In this module, we focus on a rotating frame of reference. The rotation velocity Ω represents the change in angle per time, usually expressed in rad s^{-1}. In general, it is a vector $\boldsymbol{\Omega}$ with its direction perpendicular to the plane of rotation (upward in counter-clockwise rotation). On a mass m with a radial distance r from the center of rotation, a *centrifugal force*

$$\boldsymbol{F} = m\omega^2\,\boldsymbol{r} \tag{0.1}$$

pulls the mass radially outward (think of a stone at a string).

Another inertial force in a rotating frame of reference is the *Coriolis force*

$$\boldsymbol{F} = -2m\,(\boldsymbol{\Omega} \times \boldsymbol{v}) \tag{0.2}$$

occurring when a mass moves relative to the rotating frame of reference with the velocity $\boldsymbol{v}$. Please note the vectorial notion with bold letters for the vectors. The Coriolis force causes, e.g., the low-pressure systems in the atmosphere to spin counter-clockwise in the northern and clockwise in the southern hemisphere. An important sensor example based on the Coriolis force is the gyroscope. Also, some flow sensor makes use of this force.

Sensors

The BHI260AP is a smart sensor comprising an *inertial measurement unit* (IMU) and a *microcontroller unit* (MCU). The MCU is a proprietary core called Fuser2, dedicated to sensor fusion and self-learning *artificial intelligence* (AI). Figure 5 illustrates the role of the BHI260 as a sensor hub combining the data from the internal and optional external sensors. See Figure 2 to understand the BHI260's role as a sensor hub on the Nicla Sense ME board. The BHI260 runs firmware based on Bosch Sensortec’s BSX Sensor Fusion library.

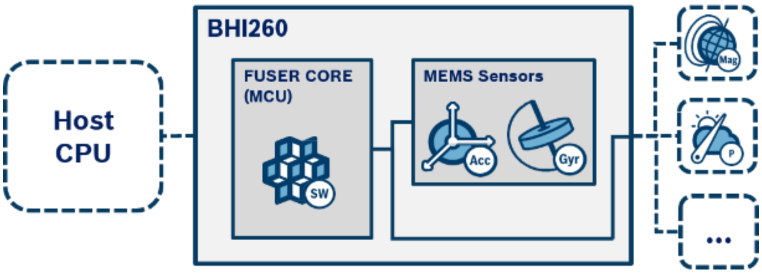


Figure 5: Block diagram of the BHI260 smart sensor with different sensors and a microcontroller. [5]

The IMU consists of gyroscopes and accelerometers for all three axes – so, in total, a 6-axis IMU with six *degrees of freedom* (DoF). Other IMUs are comprised of magnetic sensors for all three axes, in total, a 9-axis IMU. Combining the different information from six or nine axes by a *sensor fusion* algorithm provides spatial orientation. The BHI260 also allows sensor fusion with external sensors, such as the BMM150 magnetometer on the Nicla Sense ME board, resulting in a 9-axis arrangement. Beyond the spatial orientation, self-learning AI allows gesture or activity recognition. Within the lab course modules, we will not use such high-level features as the focus is on the sensor elements and principles, though you are encouraged to explore the system’s AI features on your own.

Accelerometer

An accelerometer (acceleration sensor) measures the acceleration by observing the force acting on a mass. Microfabricated accelerometers typically use the setup shown in Figure 6.

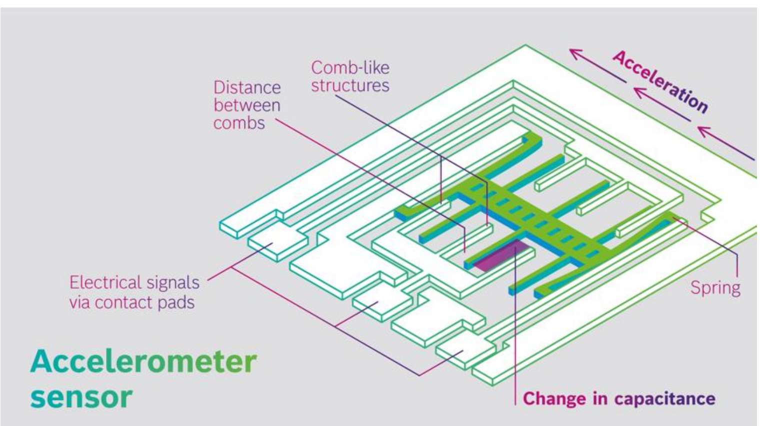


Figure 6: Typical microfabricated acceleration sensor with a moveable mass to detect the acceleration in one direction. Image: Bosch Sensortec.

Springs hold the movable mass (solid green), allowing movement along the longer axis of the mass. Comb-like structures allow the reading of the position of the mass. The mass movement causes the capacity between neighboring fingers to change (indicated by the purple plane). Practically, an electrostatic force generated from the same fingers helps to keep the mass steady by an electrical potential controlled according to the capacity values. The control signal used to keep the mass steady is the measured quantity resembling the acceleration. More details on the design and operation principle of microfabricated acceleration sensors will be provided in the lectures.

Gyroscope

The gyroscope (yaw-rate sensor) measures the rotation rate (yaw rate) by exploiting the Coriolis effect. Linear movement of a mass causes a Coriolis force perpendicular to the movement when the whole system rotates.

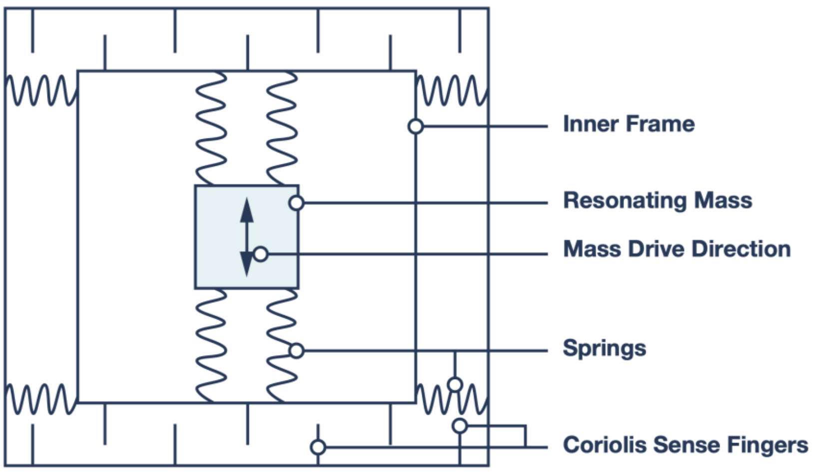


Figure 7: Typical microfabricated gyroscope (yaw-rate sensor) with an oscillating mass to detect the rotation. [6]

In a microfabricated realization of a gyroscope (Figure 7), the sensor mass, held by springs, oscillates in an inner frame (the velocity vector $\boldsymbol{v}$ is up and down in the drawing). This inner frame, held by springs, can move inside a fixed outer frame (left and right in the figure).

Rotation of the whole sensor (rotation rate vector $\boldsymbol{\Omega}$ into or out of the drawing plane) causes a Coriolis force according to the cross product in eq. 0.2. This force leads to a movement of the inner frame to the left or right in the drawing. The “Coriolis Sense Fingers” on the outer and inner frame form a capacitor. Measurement of the capacity provides the rotation rate.

Figure 8 illustrates the action of the Coriolis force for a rotation rate vector $\boldsymbol{\Omega}$ into the drawing plane. When the sensor mass moves up, the Coriolis force displaces the inner frame to the left. The displacement is onto the other side for the sensor mass moving down.

More details on the design and operation principle of microfabricated gyroscopes will be provided in the lectures.

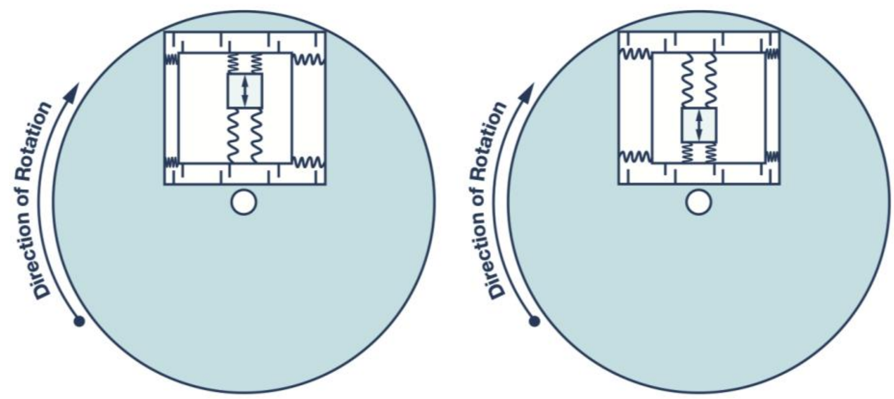


Figure 8: Illustration of the working principle for the microfabricated gyroscope from Figure 7. [6]

Sensors in the BHI260AP

The BHI260AP features acceleration sensors and gyroscopes for all three directions. Figure 9 shows the definition of the axes with the dot on the package as the reference. The positioning and orientation of the chip on the Nicla Sense ME board are clarified in Figure 10.

In the lab course, you should use the virtual sensors PASSTHROUGH, which provides the values from the physical driver directly:

```
SensorXYZ accel(SENSOR_ID_ACC_PASS);
SensorXYZ gyro(SENSOR_ID_GYRO_PASS);
```

The acceleration sensor's default configuration is the 8*g* range. The sensitivity is **4096 LSB/*g*** (least significant bit divided by free-fall acceleration), according to table 117 in [5]. To obtain the acceleration, you must divide the raw value by 4096 and multiply by 9.81 m s^{-2}.

For the rotation rate, the sensitivity is **16.4 LSB (° s^{-1}) $^{-1}$** (table 119 in [5]). Make sure not to mess up ° s^{-1} and rad s^{-1}.

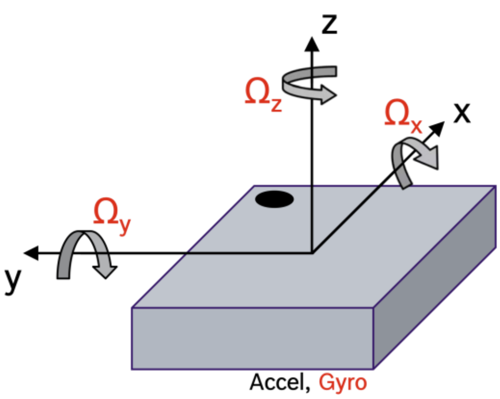


Figure 9: Orientation of the sensing axes with the dot on the BHI260AP package as the reference. [5]

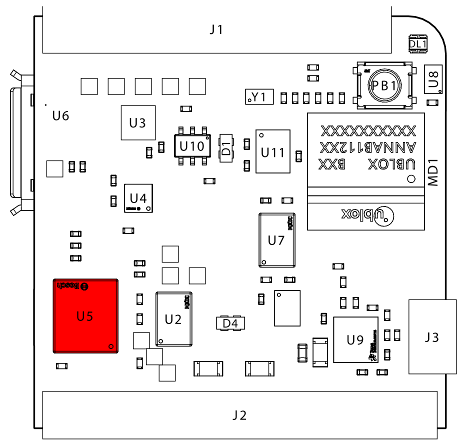


Figure 10: Top view of the Nicla Sense ME board (USB port is on the bottom) with the BHI260AP marked in red. Adapted from [1].

Getting started

An excellent introduction and source of information beyond the first steps is the Arduino Nicla Sense ME Cheat Sheet[15]. Here you can follow the links to install the IDE, get the LED blinking and learn how to communicate with the sensor in standalone mode (the Nicla Sense ME board connected to a computer with a USB cable).

Always handle the Nicla Sense ME board with care. Strictly avoid contact with liquids! Although the assembled board is not very critical concerning electrostatic discharge (ESD), you should hold the board at its edges and not tap on the surface, reducing the risk of leakage currents due to contamination.

When you are familiar with microcontroller programming, the following steps are easy. If you are new, take your time, follow the steps and optionally browse further documentation on the Arduino programming environment.

1. Install the Arduino IDE and files for the Nicla Sense ME board as described in the Cheat Sheet or using the Quickstart Guide from the board's documentation[16]. On a modern computer, we recommend the Arduino IDE 2.0. If you have older hardware and face problems with the interface's performance, you could also use the Arduino IDE 1.8.
2. Try to get the LED blinking (section "RGB LED" in the Cheat Sheet). In contrast to other boards, you can even play around with color.
3. Follow the description in the Cheat Sheet's section on "Sensors" to get temperature readings available on the serial port of your USB connection. You need to install the libraries Arduino_BHY2 and ArduinoBLE. Use the Arduino IDE's built-in serial console or any other serial terminal software to retrieve the data.

Once your setup is up and running, you are ready to start with the first tasks.

Tasks

Gyroscope

1. Write a program that measures every 100 ms the angular velocity in all three directions, and prints the values continuously on the serial interface.
    a. Ensure steady conditions for your sensor and record 1,000 gyroscope readings. Can you observe any drift? Repeat the measurements to ensure representative conditions.
    b. Calculate for each direction the mean value $\overline{\Omega}$. What is the offset in all three directions (sensor reading at rest)? Do all three directions behave the same? What could be a reason for the behavior?

[15] https://docs.arduino.cc/tutorials/nicla-sense-me/cheat-sheet
[16] https://docs.arduino.cc/hardware/nicla-sense-me

c. Show your data with subtracted mean value $\Omega - \overline{\Omega}$ for all three directions in an appropriate histogram. How are the values distributed? Do all three directions behave the same? What could be a reason for the behavior?

Merry-go-round measurements

While the sample report contains data from a real merry-go-round, you, for the sake of onboarding, can hold the sensor (and preferably your computer) in your hand and turn around yourself with bent and extended arms. There is not even need for a complete turn.

2. Write a program that measures the rotation rate in the *z*-direction and the acceleration in the *x*- and *y*-direction every 100 ms and prints the values continuously on the serial interface. Additionally, include the time output since the controller start as a timestamp for each measurement.

   a. Place the Nicla Sense ME on a merry-go-round at the outer edge of the turning part (or use your extended arm). Position the *x*-direction towards the center. Measure the distance from the board (ideally the acceleration sensor) to the center of rotation.

   b. Spin the merry-go-round (or turn around yourself) while measuring and let the rotation decay.

   c. Plot the decaying rotation rate over time. What mathematical relationship does the decay follow? Use an appropriate fit function to describe the decay.

   d. Place the sensor board closer to the center (or use your bent arm). Position the *x*-direction towards the center and measure again the distance from the center of rotation.

   e. Spin the merry-go-round (or turn around yourself) while measuring.

   f. What were the maximal rotation rates in your rides from 2c and 2d? Consider also the noise in your evaluation.

   g. What were the acceleration values in radial and tangential direction at the moments of maximal rotation rates in your rides from 2c and 2d? Discuss your results and compare your values of the radial acceleration with the acceleration based on the centrifugal force (eq. 0.1).

Module 1: Acceleration and Pressure Sensors

Acceleration sensors and navigation applications go hand in hand, while pressure sensors are probably not your first thought. At least, if you are not an experienced mountaineer or paraglider pilot, for whom still today, altitude measurements based on barometric sensors complement the navigation with the *Global Positioning System* (GPS). In this module, you use a high-resolution pressure sensor to characterize an elevator ride and compare the data with readings from the acceleration sensor.

Barometric Formula

The atmospheric pressure depends on temperature and the weather conditions but also the altitude. For a period short enough to neglect the influence of weather change, the pressure changes at 250 m altitude approximately by 11.3 Pa m^{-1} (or 10 Pa m^{-1} if you prefer a rule of thumb).

You might remember a better description from your physics class, the barometric formula. For small changes in altitude, the influence of the temperature change with height is negligible (isothermal atmosphere), and the pressure p relates to the height above sea level h exponentially

$$p = p_0 \exp\left(-\frac{gMh}{RT}\right), \tag{1.1}$$

where p_0 is the reference pressure, and T is the absolute temperature. The constants are standard gravity $g = 9.81$ m s^{-2}, the molar mass of air $M = 0.02896$ kg mol^{-1}, and the universal gas constant $R = 8.314$ J mol^{-1} K^{-1}.

Sensors

This module uses both the accelerometer from the BHI260AP and the pressure sensor. See Module 0 for the description of the acceleration sensor.

Pressure Sensor

Microfabricated pressure sensors are based on a membrane deflection caused by a pressure difference between the two sides of the membrane. In a differential pressure sensor, both sides of the membrane are accessible from the outside, unlike an absolute pressure sensor (barometric pressure sensor) with the membrane sealing a cavity with a known reference pressure. The BMP390 is an absolute pressure sensor that matches the atmospheric pressure range even at high altitudes.

Figure 11 shows the typical setup of a microfabricated pressure sensor. A higher pressure above the silicon membrane causes a deflection which piezoresistive elements connected in a bridge circuit detect as a voltage change. A critical issue of such a semiconductor pressure sensor is its temperature dependency. That is why more accurate sensors include a temperature sensor for compensation with digital signal processing.

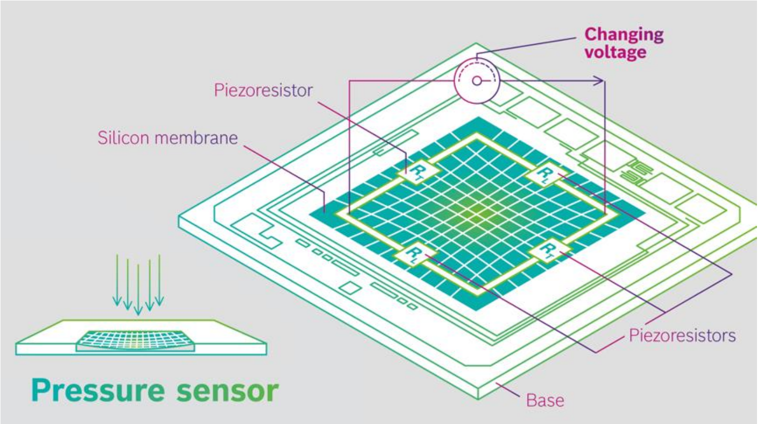


Figure 11: Typical microfabricated pressure sensor with piezoresistive elements measuring the deflection of a membrane. In an absolute pressure sensor, the membrane seals a cavity with the reference pressure. Image: Bosch Sensortec.

BMP390

The BMP390 combines a *microelectromechanical system* (MEMS) sensor element with an *application-specific circuit* (ASIC). The ASIC features the analog front-end and a 24-bit *analog-digital converter* (ADC) together with logic for compensation purposes (see Figure 12 for the block diagram). The sensor features a relative accuracy of ±0.03 hPa, translating into a relative accuracy better than ±0.3 m in a height measurement (see eq. 1.1).

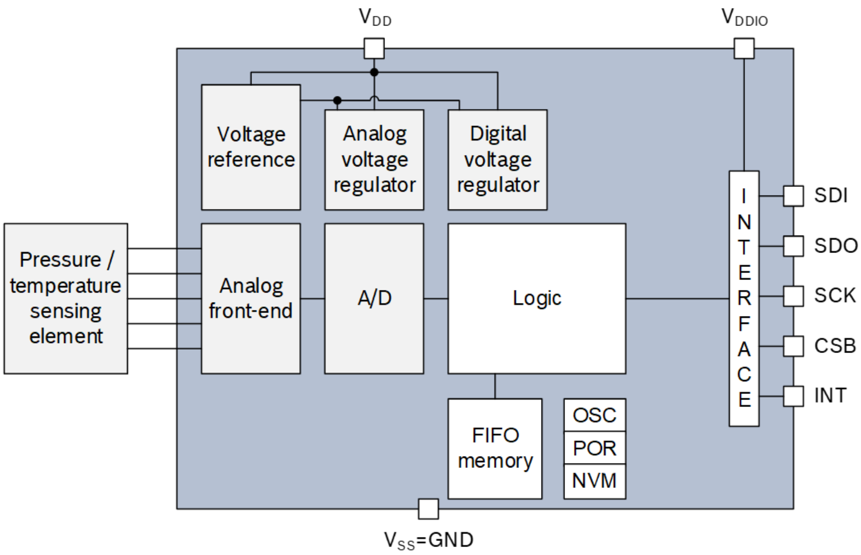


Figure 12: Block diagram of the BMP390. [7]

The metal case of the sensor has a tiny hole to allow the pressure exchange (Figure 13 left). Can you see it with bare eyes on your Nicla Sense ME board? The teardown of the BMP390 (Figure 13 right) shows two separate dies: the MEMS sensor element with the membrane placed on the ASIC and connected by five bond wires. With a closer look at the membrane, you can even guess the location of the piezoresistive elements.

Figure 13: BMP390 in its housing with the tiny opening at the top for pressure exchange (left) [6] and the sensor with the cap removed (right) [8].

In the lab course, you should use the virtual sensors SENSOR_ID_BARO, which provides the values from in the BMP390:

```
Sensor baro(SENSOR_ID_BARO);
```

The obtained values are the pressure in hPa (100 Pascal) to match the traditionally used unit millibar (mbar).

Tasks

Pressure Sensor

1. Write a program that measures every 100 ms the pressure and prints the values continuously on the serial interface.
    a. Ensure steady conditions for your sensor and record 1,000 pressure readings. Can you observe any drift? Repeat the measurements to ensure representative conditions.
    b. Convert the raw readings to pressure. Calculate the mean value $\bar{p}$. What is the offset (sensor reading at rest)? Is this value expected?
    c. Show your data with subtracted mean value $p - \bar{p}$ in an appropriate histogram. How are the values distributed? How do your results fit to the datasheet?

Acceleration Sensor

2. Write a program that measures every 100 ms the acceleration in all three directions and prints the values continuously on the serial interface.
    a. Ensure steady conditions for your sensor and record 1,000 acceleration readings. Can you observe any drift? Repeat the measurements to ensure representative conditions.

b. Calculate for each direction the mean value $\bar{a}$. What is the offset in all three directions (sensor reading at rest)? Do all three directions behave the same? What could be a reason for the behavior?

c. Show your data with subtracted mean value $a - \bar{a}$ for all three directions in an appropriate histogram. How are the values distributed? Do all three directions behave the same? What could be a reason for the behavior?

Elevator Measurement

3. Write a program that measures the pressure and the acceleration in z-direction every 100 ms and prints the values continuously on the serial interface. Additionally, include the time output since the controller start as a timestamp for each measurement.

   a. Place your setup in an elevator (e.g., the elevator in lecture building 101). Make sure that your board is parallel to the floor and does not move during the ride. Start your measurement before the elevator moves. Ride the elevator from the ground floor to another floor and back. Compare different the different directions (upwards/downwards). Please ensure not to disturb other people.

   b. Evaluate the pressure data using the barometric formula (eq. 1.1) to describe the height of your rides in a diagram (height vs. time).

   c. Integrate the acceleration sensor data to show the velocity of your rides in a diagram (velocity vs. time).

   d. Integrate the acceleration sensor twice to show the height of your rides in a diagram (height vs. time) together with readings from the pressure sensor. Discuss which approach suits better if you want to track the height during your elevator rides.

Module 2: Magnetic Sensors

In contrast to the previous modules with acceleration, rotation, and pressure, we focus here on a quantity, the magnetic field intensity, which humans hardly can sense. While some animals can orient themselves according to the earth's magnetic field, we need technical help. For a long time, people have been using a pivoted magnet in the form of a needle to figure out the north-south axis. In the Chinese Han Dynasty, more than 2000 years ago, the compass was one of the Four Great Inventions. In Europe, the compass arrived in the 12th century and enhanced astronomic navigation. The measurement of the north direction became an essential tool for medieval seamen and arguable was one of the critical factors in European exploration and later, unfortunately also, lead to the colonialization of large parts of the world.

We will use the magnetic field sensor of the Nicla Sense ME board to characterize the earth's magnetic field (Figure 14) and evaluate how well we can find the magnetic north pole using the sensor's raw values. As we are all used to relying on the Global Positioning System (GPS) or even navigation based on GPS fused with cellular data on our smartphones, this module may inspire you to explore the traditional way.

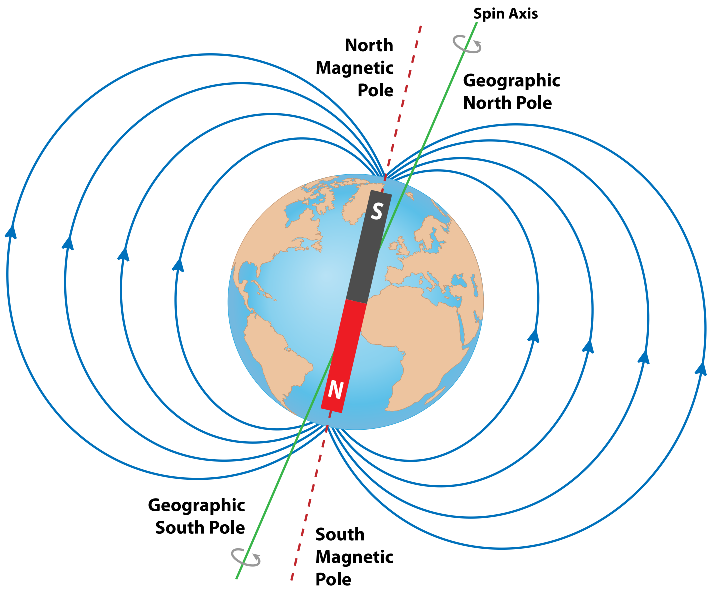


Figure 14: The earth's magnetic field forming the magnetic poles has an axis deviating slightly from the spin axis. The north magnetic pole (to which the north pole of a pivoted magnetic needle points) is a magnetic south pole.

Earth's Magnetic Field

Although we cannot sense it, we all depend on the earth's magnetic field vitally. Without its shielding properties, we would be exposed to solar winds (streams of charged particles emitted by the sun). Polar lights are an appealing side effect of this shielding effect. While one might think of the earth as a huge permanent magnet (as suggested by the illustration in Figure 14), a closer look reveals its impossibility. At a depth of 20 to 30 km already, the temperature is above iron's *Curie temperature* (the temperature at which a material loses its permanent magnetic properties). The commonly accepted *dynamo theory* assumes movement in the earth's liquid inner core, inducing currents in the solid outer core, causing the earth's magnetic field.

The inner core's flow, including its direction, depends on many factors only loosely connected to the earth's rotation. Accordingly, the magnetic north-south axis deviates from the earth's rotation axis. Because of inhomogeneous flow conditions, regional anomalies exist.

In Germany, the *declination*, the deviation of the compass needle from magnetic north is only up to 3°. Other regions are more critical, having made medieval seamen sometimes struggle. That is, for example, why we are still not sure at which position Christopher Columbus landed in America. Additionally, ores in the outer core influence the magnetic field locally. For example, in northern Sweden, in Kiruna, there are places where the compass needle would point to the south.

Figure 15: Wandering of the north magnetic pole within the last centuries.

The exact position of the *north magnetic pole* (the point at which a compass needle would rotate freely) also changes with time. Figure 15 shows the trace since 1590. In this figure, you also find a point labeled the *north geomagnetic pole*. This point refers to an ideal dipole model fitting the Earth's magnetic field. More advanced models account for regional anomalies too. The International Geomagnetic Reference Field (IGRF) and the World Magnetic Model (WMM) are two commonly used magnetic earth models[17] to describe the field at a specific location and date. For reference data needed in the lab course, both models are fine.

Quantitative Description

The earth's magnetic field is described at specific locations by the horizontal, vertical, and total field intensity. The different parameters relate to sensor readings by simple vector calculations:

- Horizontal magnetic field intensity $B_{\mathrm{hor}} = \sqrt{B_x^2 + B_y^2}$
- Vertical magnetic field intensity $B_{\mathrm{vert}} = B_z$
- Total magnetic field intensity $B = \sqrt{B_x^2 + B_y^2 + B_z^2}$.

The coordinate system assumes the *z*-axis to be perpendicular to the ground.

[17] You can use an online tool (e.g., https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml) or a mobile phone app to get the values for your location.

Sensors

Hall Sensor

The most widely known magnetometer is the Hall sensor. The principle exploits the *Hall effect*, the formation of a voltage perpendicular to a current flow when charge carriers get deflected by a magnetic field. The conductor in a Hall sensor typically has the geometry of a thin plate, either with longer dimensions along the direction of current flow or, more commonly nowadays, a quadratic footprint.

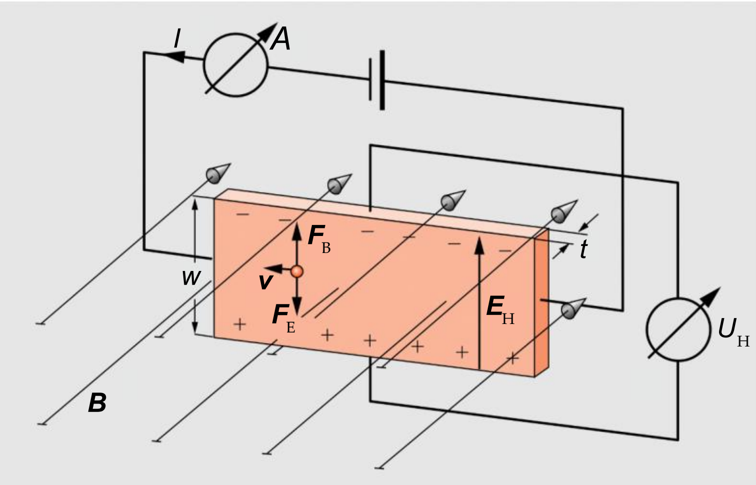


Figure 16: Hall effect: the magnetic field (intensity vector $\boldsymbol{B}$) causes deflection of charge carriers leading to a Hall voltage U_H across the width of the conductor. Adapted from [9].

Figure 16 illustrates the principle of the Hall effect. A magnetic field (intensity vector $\boldsymbol{B}$) points orthogonally to a conductor where charge carriers (current I, velocity vector $\boldsymbol{v}$) get deflected. An electric field (the *Hall field* with intensity vector $\boldsymbol{E}_\mathrm{H}$) establishes, causing an electric force ($\boldsymbol{F}_\mathrm{E}$) balancing the *Lorentz force* ($\boldsymbol{F}_\mathrm{L}$):

$$e\boldsymbol{E}_\mathrm{H} = -e\,\boldsymbol{v} \times \boldsymbol{B} \tag{2.1}$$

e is the elementary charge. The Hall voltage across the width w of the conductor relates to the charge carrier's velocity and the orthogonal component of the magnetic field intensity:

$$U_H = -wvB \tag{2.2}$$

Expressing the Hall voltage in terms of current I using the charge carrier density n makes clear why thin Hall plates (thickness t) are advantageous:

$$U_H = -\frac{IB}{ent} \tag{2.3}$$

More details on the Hall sensors and their technological realization in silicon will be provided in the lectures.

FlipCore Element

The FlipCore technology is a proprietary technology by Bosch Sensortec. The principle relates to a *fluxgate sensor* in which a sender coil drives a core between its saturation magnetization, and a receiver coil wound on the same core reads the induced signal. An external magnetic field (the field intensity to be measured) superimposes and causes a signal detected as 2nd harmonic.

The FlipCore element simplifies the arrangement using a thin-film core of an iron-nickel alloy, typically a stack of few separated layers with a thickness of a few ten nm only, containing a single magnetic domain each. Upon periodic excitation, the magnetization flips between its saturations. An external magnetic field superimposing the periodical excitation causes a delay at the receiving coil, allowing for much simpler time measurement than the fluxgate sensor's detection principle. The FlipCore principle features reduced power consumption and lower noise compared to a Hall sensor.

BMM150

The BMM150 contains two FlipCore elements for the in-plane magnetic field components and one Hall sensor for the vertical component. The Flip Core elements and the Hall sensor have their own circuits, which all use the same ADC (Figure 17).

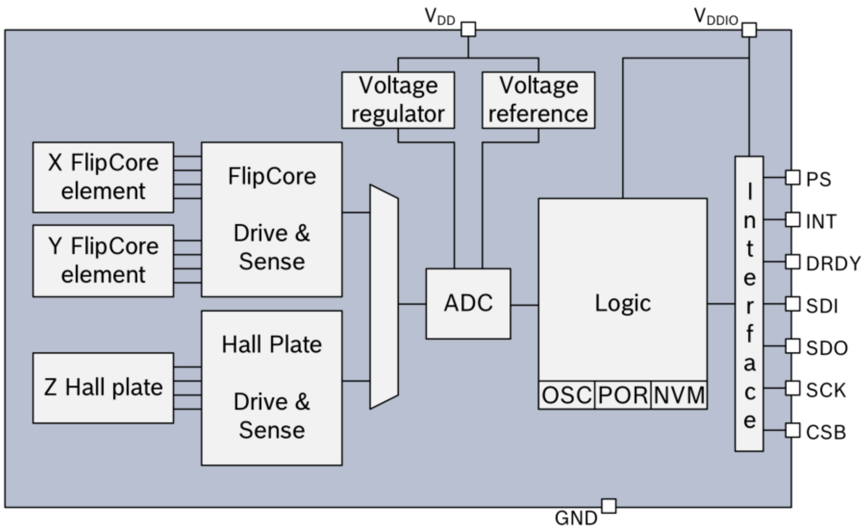


Figure 17: Block diagram of the BMM150. [10]

The inner life of the comparable BMC050 reveals the two separate FlipCore elements for the in-plane components integrated next to the stack of dies with the Hall plate below the main circuit (Figure 18).

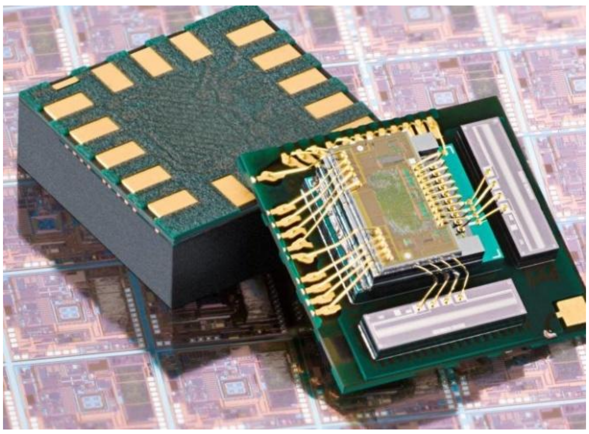


Figure 18: View into the BMC050, a comparable magnetic sensor from Bosch. [11]

While we focus in the lab course on unprocessed values from the sensor, the designer implementing a sensor in an application would prefer compensated data taking advantage of the manufacturer's knowledge of the device details. A typical application requiring sophisticated compensation is a magnetometer installed in a mobile phone. Here, sensor-specific properties play a role, but also the environment, like the field from the magnet of a loudspeaker with orders of magnitude higher strength than the magnetic field of interest, namely the earth magnetic field, on which it is superposed. Most critically, changing environmental conditions can be challenging to handle. For example, a temperature change of just 10 K could cause a difference in the speaker magnet's field intensity in the order of the entire earth's magnetic field intensity. Thus, sophisticated algorithms aim to distinguish desired effects from the environment's contribution.

In the lab course, you should use the virtual sensor PASSTHROUGH, which provides the values from the physical driver directly:

```
SensorXYZ magn(SENSOR_ID_MAG_PASS);
```

The sensitivity of the returned magnetic field intensity is **16 LSB µT^{-1}** [10].

Figure 19 shows the definition of the axes with the dot on the package as the reference. The positioning of the chip on the Nicla Sense ME board gets clear in Figure 20.

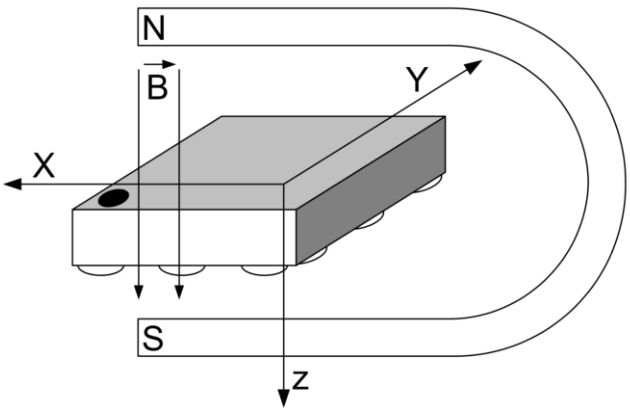


Figure 19: Orientation of the sensing axes with the dot on the BMM150 package as reference. [10]

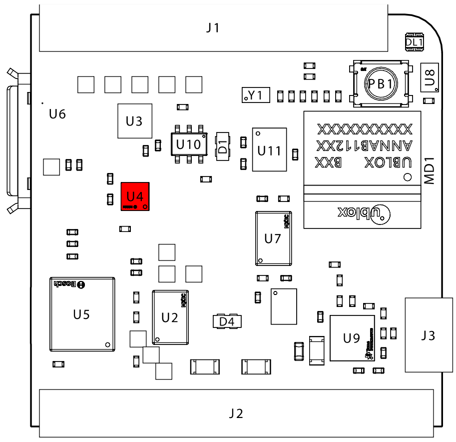


Figure 20: Top view of the Nicla Sense ME board (USB port is on bottom) with the BMM150 marked in red. Adapted from [1].

Tasks

Magnetic Sensor

1. Write a program that measures the magnetic field intensity in all three directions (B_x, B_y, B_z) every 100 ms and prints the value on the serial interface.
    a. Ensure steady conditions for your sensor and record 1,000 magnetic field readings. Can you observe any drift? Repeat the measurements to ensure representative conditions.
    b. Calculate for each direction the mean value $\bar{B}$ and show your data with subtracted mean value $B - \bar{B}$ in an appropriate histogram. How are the values distributed? Are all three directions behaving the same? Do you observe any difference between the FlipCore elements and the Hall sensor?
2. Evaluate the offset of all three axes:
    a. Ensure steady conditions for your sensor and record 100 magnetic field readings. Take the mean value in each direction.
    b. Flip your board 180° around the *x*-axis and repeat the measurement for offset correction. Then, flip your board 180° around the *y*-axis and repeat the measurement. Ensure that the board's positioning does not change or that no external factors affect the magnetic field.
    c. Compare your results from 2a and 2b for each direction. What is the sensor's offset? Does this match the range specified in the datasheet? Are all three directions behaving the same? Do you observe any difference between the FlipCore elements and the Hall sensor?

Earth's Magnetic Field

3. Use your program from the first task to measure the earth's magnetic field at your place. In your evaluation, consider the offsets obtained in task 2.
    a. Find a place outdoor where you can perform an undisturbed measurement. Avoid metals that might influence the magnetic field locally. For each direction, use the mean value of the acquired data and report it together with the standard deviation (e.g., (42 ± 3) µT).
4. Evaluate the horizontal magnetic field intensity B_{hor}, the vertical component B_{vert}, and the total intensity B.
    a. Compare your findings with the expected value according to a geomagnetic model for your location (longitude, latitude) and date.
    b. Look up the specifications of the BMM150 and discuss your results regarding offset, resolution, and accuracy.
5. Place your sensor board aligned with a building or straight street for which you know the north direction. You could measure the north direction from a satellite image, e.g., using

Google Earth Pro[18] or similar software. On the campus, you find a north-south street (from the corner behind the stone garden next to building 101, along buildings 051 and 052, towards building 105). Avoid metals that might influence the magnetic field locally.

a. Measure B_x and B_y. Use trigonometric calculation to obtain the angle with respect to the north direction.

b. How good does your measured direction match the expected north direction? Discuss your results. Are your results accurate enough to relate possible deviations to the declination?

[18] https://www.google.com/earth/about/versions/#download-pro

Module 3: Gas Sensors

We usually pay attention to air quality only when it is not good anymore. Especially for outdoor air quality, we realize the comfortable situation around Freiburg only when we come back from other regions worldwide where pollution is sometimes so severe that we can hardly ignore it. On a longer timescale, elevated levels of particulate matter, nitrogen oxides (NO_x), and sulfur dioxide (SO_2) cause severe health problems and have a substantial death toll.

In principle, all outdoor pollutants may play a role indoors too, but, especially in first-world countries, the outdoor air quality is satisfactory in most areas. Here, humans exhaling carbon dioxide (CO_2) or *volatile organic compounds* (VOC) cause “bad air.” Artificially generated VOCs released from materials may add on. While air purifiers can filter out VOCs, ventilation is the simple answer to improve indoor air quality, including lowering CO_2 concentration to atmospheric levels. Since the COVID-19 pandemic, the interest in indoor air quality has grown, using it as a marker for potentially infectious aerosols exhaled by humans.

In this module, we use the Nicla Sense ME metal oxide gas sensor to assess indoor air quality and explore how rather unspecific features allow for estimating meaningful parameters (concentrations of VOCs or CO_2).

Indoor Air Quality

The concept of air quality is complex and mainly focused on the absence of pollutants. But also, humidity and temperature play an essential role.

Composition of Air

The earth’s atmosphere contains mainly nitrogen (N_2, around 78%) and, for us, most importantly, oxygen (O_2, around 21%). Table 1 summarizes the typical composition of the atmosphere. The CO_2 concentration changes dramatically (Figure 3), e.g., the U.S. Standard Atmosphere dating back to 1976, which considered CO_2 in a fraction of 314 ppm only [12].

Table 1: Composition of the atmosphere (dry air at sea level) considering the top five gases only; the mole and volume fractions are the same, assuming ideal gas.

Gas	Mole fraction, volume fraction
Nitrogen	78.08%
Oxygen	20.95%
Argon	0.93%
Carbon dioxide	415 ppm (October 2022)
Neon	18 ppm

Volatile Organic Compounds

Organic compounds are molecules containing carbon-hydrogen or carbon-carbon bonds. High vapor pressure at room temperature classifies a *volatile organic compound (VOC)*. Per definition, methane is often not considered a VOC. Typical VOCs are benzene or ethanol; common biogenic VOCs are, e.g., isoprene or terpenes.

Carbon dioxide

When humans inhale air into the lungs, oxygen gets into the blood, and the body releases CO_2 instead, lowering the oxygen concentration in exhaled air to 17%. The approximately 4% of carbon dioxide in the breath increases the CO_2 concentration in indoor scenarios if the environment is not well-ventilated. The optimal concentration is below 800 ppm, and a typically recommended threshold for ventilation is 1000 pm. In Germany, the occupational exposure limit is 5000 ppm. However, adverse effects on our ability to concentrate are already around 2000 ppm. Above 1% of CO_2, symptoms like drowsiness and headache set in, and a concentration of 8% causes death in less than an hour.

Indoor Air Quality Index

The indoor air quality (IAQ) index is a measure of air quality focused on humans, a number typically between 0 and 500 (0 is the best). Worldwide different definitions exist, considering various parameters to define the index. Wikipedia[19] provides a good summary of the different definitions. The BME688 used in the lab course provides a scale illustrated in Figure 21. The values are calculated based on the sensor readings using a proprietary algorithm.

IAQ Index	Air Quality	Impact (long-term exposure)	Suggested action
0 – 50	Excellent	Pure air; best for well-being	No measures needed
51 – 100	Good	No irritation or impact on well-being	No measures needed
101 – 150	Lightly polluted	Reduction of well-being possible	Ventilation suggested
151 – 200	Moderately polluted	More significant irritation possible	Increase ventilation with clean air
201 – 250[9]	Heavily polluted	Exposition might lead to effects like headache depending on type of VOCs	optimize ventilation
251 – 350	Severely polluted	More severe health issue possible if harmful VOC present	Contamination should be identified if level is reached even w/o presence of people; maximize ventilation & reduce attendance
> 351	Extremely polluted	Headaches, additional neurotoxic effects possible	Contamination needs to be identified; avoid presence in room and maximize ventilation

Figure 21: Indoor Air Quality (IAQ) Index as returned by the BME688. [13]

Sensors

Metal Oxide Gas Sensors

Metal oxide (MOX) gas sensors comprise a semiconducting oxide layer at which gases get adsorbed and oxidized or reduced at elevated temperatures, altering the MOX's resistance (Figure 22). Typically, the gas sensing layer(s) are deposited on a hotplate comprising a

[19] https://en.wikipedia.org/wiki/Air_quality_index

heating element, e.g., based on platinum. Microfabrication enables thin heater membranes, allowing for a fast operating temperature change.

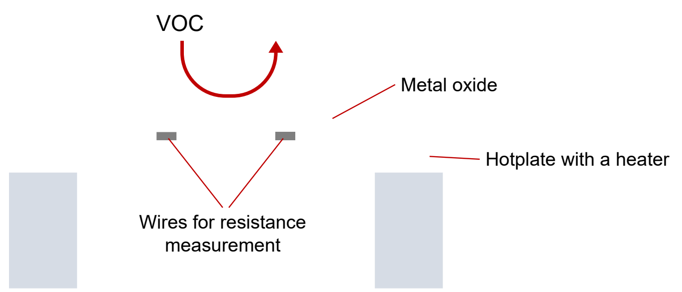
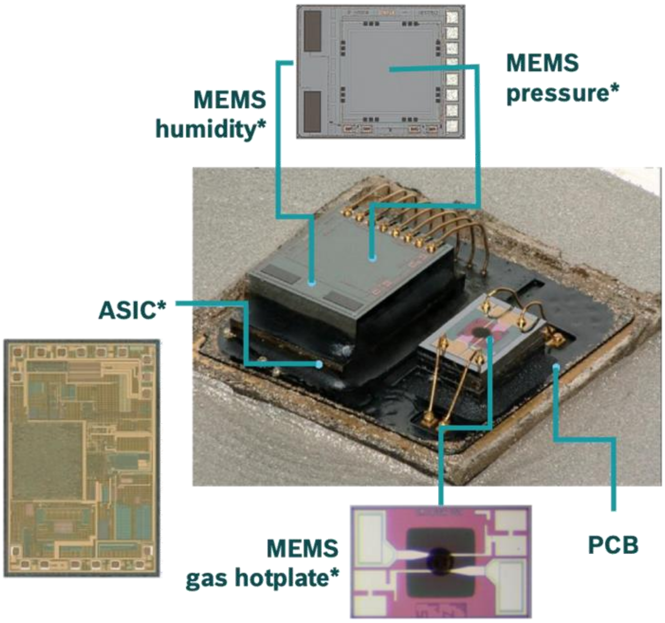


Figure 22: Principle of a metal oxide (MOX) gas sensor, the sensing element is on a hotplate heated to several hundred °C at which gases are oxidized or reduced, changing the layer's resistivity.

At high temperatures, the adsorbates are removed, while at the operating temperature, different gases contribute to the MOX layer's resistivity by reacting with oxygen atoms at the surface. Reducing gases (e.g., VOCs) lower the resistance while oxidizing gases (e.g., NO_x compounds) cause an increase in the resistance. Additionally, water molecules (humidity) play a role why often MOX gas sensors come along with a humidity-sensing element for correction purposes. While MOX gas sensors are very sensitive, selectivity is poor. Apart from the material itself, the operating temperature allows for some selectivity.

BME688

The BME688 features a MOX gas sensor together with sensors for temperature, humidity, and pressure. Figure 23 shows the inner life of the closely related BME680. The left stack comprises the *application-specific integrated circuit* (ASIC) below and the humidity, temperature, and pressure sensors on top. The smaller part on the right is the MOX gas sensor itself. In the top view of the gas sensing element (lower inset), you can see the hotplate (dark square) with the MOX on top (black round spot).



* *Picture source: System Plus Consulting | BME680*

Figure 23: BME680 gas sensor, which is closely related to the BME688 used in the lab course. Image: Bosch Sensortec/System Plus Consulting.

A detailed look at the gas-sensing element by *scanning electron microscopy* (SEM) reveals the rough and partially crystalline nature of the metal oxide placed on the two metal lines for resistance measurement (Figure 24).

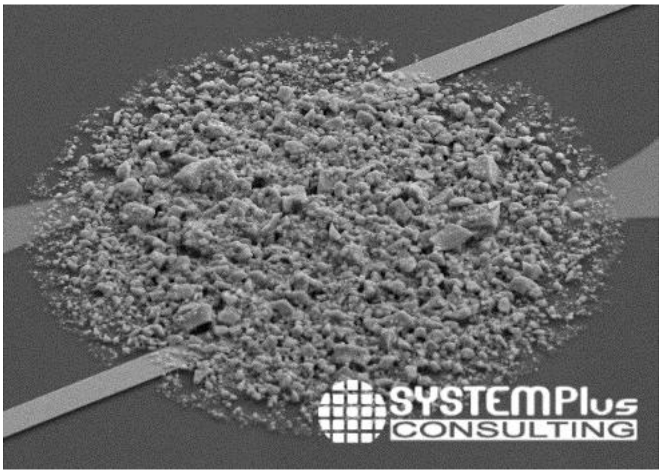


Figure 24: SEM image of the gas sensing element of a BME680 [14].

The gas sensor allows for direct readout of the MOX. The virtual sensor SENSOR_ID_GAS, provides the resistance value directly:

```
Sensor gas(SENSOR_ID_GAS);
```

The function gas.value() returns the resistance in Ohms.

However, the usual mode of operation uses the onboard gas scanner functionality, which applies a sophisticated temperature cycle to the metal oxide (Figure 25). The sensing element is sensitive to VOCs but also reacts to other gases such as sulfur compounds, carbon monoxide (CO), and hydrogen (H_2). The BME688 uses the *Bosch Software Environmental Cluster*

(BSEC) library to estimate different gas sensing parameters based on several features, which are resistance readings at different time points in the temperature cycle.

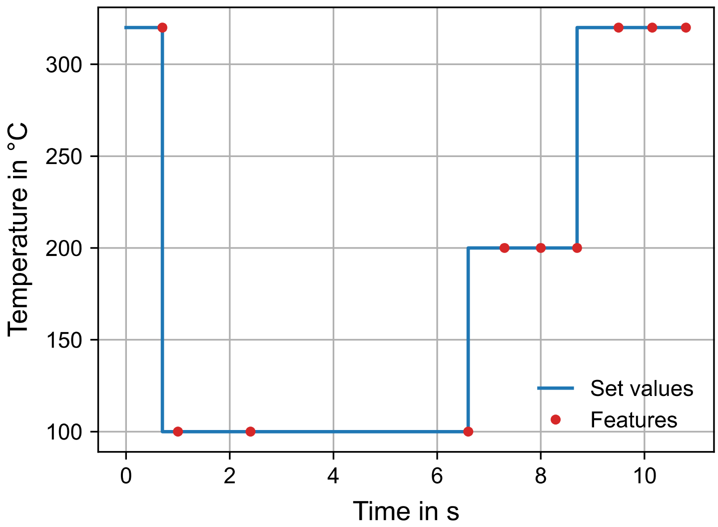


Figure 25: Temperature profile ("standard scan") used in the BME688. Adapted from [13].

The BSEC allows the generation of custom temperature cycles compromising between higher selectivity, lower energy consumption, and faster acquisition rates. The functionalities of the Nicla Sens ME board are limited to the cycle shown in Figure 25. You can operate the BME688 with the BSEC functionality using the virtual sensor SENSOR_ID_BSEC:

```
SensorBSEC bsec(SENSOR_ID_BSEC);
```

The BSEC algorithm provides values for the IAQ (bsec.iaq()), the VOC concentration assuming human breath as the main source (bsec.b_voc_eq()), and even the CO_2 concentration (bsec.co2_eq()), the latter two in ppm. While the MOX sensor shows no sensitivity to CO_2, the algorithm uses different features to estimate the CO_2 concentration, assuming humans as the major source of the detected VOCs. Additionally, the BSEC library provides the temperature (bsec.comp_t()) and the humidity (bsec.comp_h()).

The BSEC algorithm continuously recalibrates its estimation. The bsec.accuracy() function returns different levels:

- Level 0: not calibrated
- Level 1: calibration started
- Level 2: calibration ongoing, target of error not reached
- Level 3: calibration finished

Please note that it may take hours until the algorithm reaches level 3. This process takes much less time when several "strong events" occur, maybe even slightly blowing toward the sensor.

Tasks

MOX sensor raw values

1. Write a program that reads the resistance of the BME688 gas every 10 s and prints the value on the serial interface. Let your program run for at least 10 minutes before the measurements.
    a. Explore different sources of VOCs in your environment. Interesting examples are a package of tea leaves or coffee beans and the smell of different spices. You can even detect the ethanol used as a preservative in ready-made dough products such as pizza or frozen bread rolls once you start heating them to slightly above room temperature.
    b. Document the resistance value of at least three examples. Is the value constant over time? How fast does the value recover once the source is removed?

BSEC library

2. Write a program that reads the parameters for VOC and CO_2 concentration together with the accuracy level of the BME688 using the BSEC functionality every 10 s and prints the value on the serial interface.
    a. Wait until an accuracy level of 3 is reached at least once.
    b. Document the concentrations value of the examples from task 1b. Are the values constant over time? How fast do the values recover once the source is removed?
    c. Try to find an example from which it is evident that the BME688 does not detect CO_2 directly but estimates its concentration from other parameters.
3. Enhance your program to measure the IAQ, temperature, and humidity.
    a. Wait until an accuracy level of 3 is reached at least once.
    b. Monitor a typical indoor situation (student home, kitchen in a shared flat, a room in which a party happens, etc.) over at least 24 hours. Note all remarkable events. Please ensure that it is fine for all involved persons that you monitor the environment with the BME688 and share the data in the lab report. Please do not forget to disable the power-save features of your computer.
    c. Report your results as a time trace with daytime on the *x*-axis. Thoroughly discuss your data. How would you consider the air quality during your measurements? How realistic are the estimates from the BSEC algorithm?

Recommended Literature

- Arduino Nicla Sense ME Cheat Sheet: https://docs.arduino.cc/tutorials/nicla-sense-me/cheat-sheet

Depending on your preferred language:

- Meschede, Gerthsen: „Gerthsen Physik“, 2015, Springer Spektrum (Campus-Lizenz: http://www.redi-bw.de/start/unifr/EBooks-springer/10.1007/978-3-662-45977-5)
- Halliday, Resnick, Walker: “Fundamentals of Physics”, 2010, Wiley

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[8] System Plus Consulting https://s3.i-micronews.com/uploads/2017/11/Yole_Bosch_Sensortec_BMP380_barometric_pressure_sensor_flyer_SP17358.pdf [Accessed: 18-Aug-2022].

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